

Analysis

2024-10-10

```
library(ggplot2)
```

Including zero results

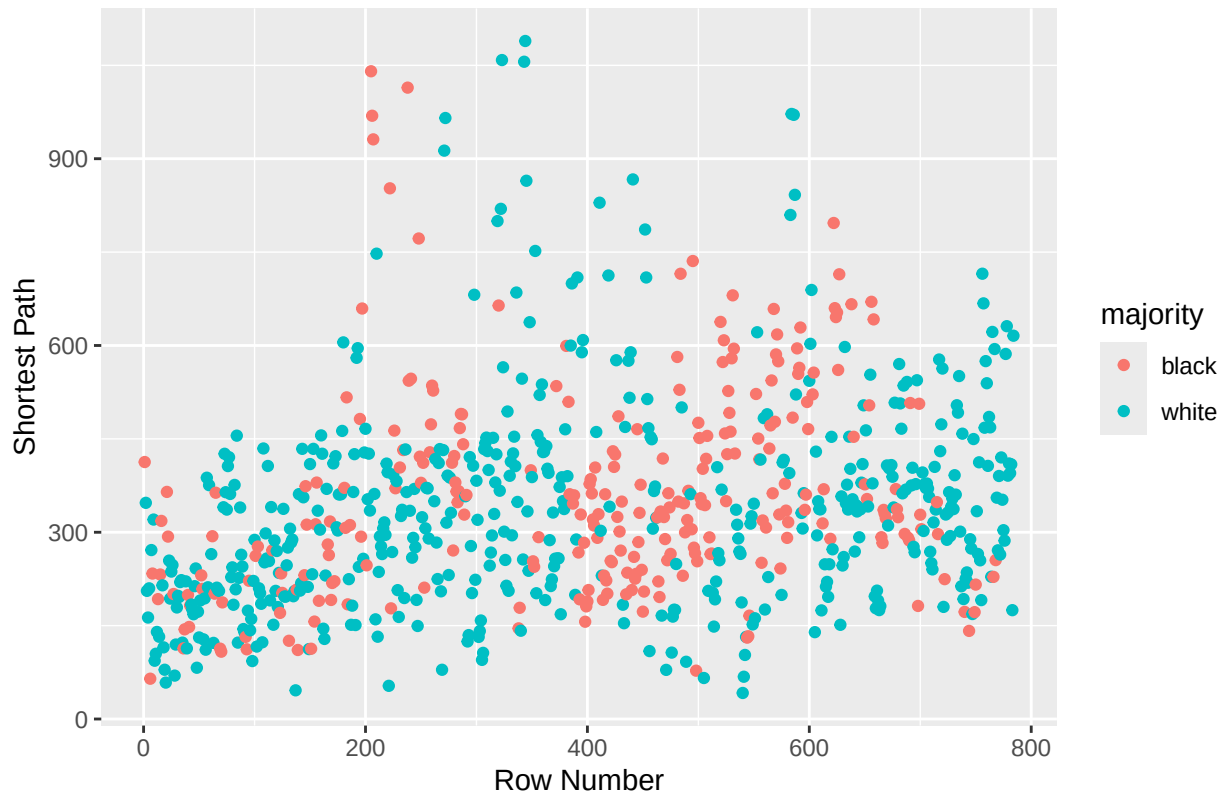
```
data <- read.csv('./hospital-3-closest.csv')
```

```
head(data)
```

```
##           X majority shortest_path shortest_restricted_path
## 1 Census Tract 8424    black      412.76667             418.76667
## 2 Census Tract 8403    white      347.26667             348.00000
## 3 Census Tract 8411    white      205.83333             205.83333
## 4 Census Tract 8412    white      163.03333             163.03333
## 5 Census Tract 8390    white      209.93333             209.93333
## 6 Census Tract 8382    black       64.73333              64.73333
##  relative_increase absolute_increase n_cameras
## 1          0.01453606          6.0000000 1.0000000
## 2          0.00211173          0.7333333 0.6666667
## 3          0.00000000          0.0000000 0.0000000
## 4          0.00000000          0.0000000 0.0000000
## 5          0.00000000          0.0000000 0.0000000
## 6          0.00000000          0.0000000 0.0000000
```

```
ggplot(data, aes(x = seq_along(shortest_path), y = shortest_path, color = majority)) + geom_point() +
  labs(x = "Row Number", y = "Shortest Path", title = "Time to reach destination by demographic group")
```

Time to reach destination by demographic group



```
summary(lm(formula=absolute_increase ~ majority + shortest_path, data=data))
```

```
##
## Call:
## lm(formula = absolute_increase ~ majority + shortest_path, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -48.710 -14.989  -3.881   9.199 191.257
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -21.92919    2.508474  -8.742  < 2e-16 ***
## majoritywhite  16.145908    1.933464   8.351 3.08e-16 ***
## shortest_path  0.095104    0.005426  17.527  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 25.77 on 781 degrees of freedom
## Multiple R-squared:  0.3142, Adjusted R-squared:  0.3124
## F-statistic: 178.9 on 2 and 781 DF, p-value: < 2.2e-16
```